

The rise of machine intelligence

H S Nwana, N Azarmi and R Smith

This paper provides a brief historical introduction to the machine intelligence endeavour, and to the discipline of artificial intelligence (AI) whose techniques are employed in building intelligent software systems. It also presents some AI success stories both outside and within BT.

1. The machine intelligence dream

We draw from the prologue of Kurzweil's excellent book, *The Age of Intelligent Machines* [1], in order to chronicle briefly the genesis of the dream of machine intelligence.

We are currently in the thick of the second Industrial Revolution. The first arguably began when a young Englishman, John Kay, invented his new machine for weaving cloth. Though he died penniless, because he spent all his money on litigation attempting to enforce the patent he had obtained for his invention, little did he know that he had set in motion a revolution that will see the automation of all aspects of the production of cloth by the end of the eighteenth century.

Good ideas are infectious, and others in other industries observed the dramatic rise in productivity that automation had brought to English textiles. The next two centuries after Kay's invention saw the spread of mechanisation to other industries and countries including the rest of Europe, the United States and Japan. The economies of nations shifted from agrarian and craft economies to ones dominated by machines. The chain of successive improvements in automation continues, as we know, to this day, even more speedily after the invention of the transistor which ushered in the second industrial revolution. Such automation naturally presents threats, and in the context of the first industrial revolution, it prompted Ned Ludd to found the Luddite movement which set out to protect the livelihoods (as they saw it) of the textile workers. They failed, and, ever since, the 'automation versus jobs' debate has been a controversial and sensitive issue.

However, Kurzweil correctly points out that the first industrial revolution extended, multiplied and leveraged our **physical** capabilities — making us humans able to

manipulate tasks which our muscles cannot, and at speeds which were hitherto unachievable.

The second industrial revolution, the one we are currently in, is based on machines that extend, multiply and leverage our **mental** abilities. Naturally, it is accompanied too by similar economic and social controversies as the first. As *homo sapiens*, we operate with symbols and language — and we are now designing machines which manipulate symbols and languages, though generally not natural languages. These machines or computers of the second industrial revolution are based on transistor and semiconductor technology whose growth over the last three decades has, by all accounts, been phenomenal. Computer power doubles every two years and has done so for decades. Today, machines or computers pervade all aspects of our lives, from the mundane (as in billing systems within a company like BT) to the controversial (as in smart weapons which were deployed during the Gulf war).

Companies or corporations which make most use of these machines now dominate others — the Japanese 'miracle' is an obvious case in point. The use of machines is now a *sine qua non* for international and global trade, so much so that they have been blamed in some quarters for the stock market crash of 1987. While the first industrial revolution was fuelled by natural resources, the second is being fuelled by knowledge — which justifies the fourth century slogan: 'Knowledge is power'. Today, a sure way for a company to commit commercial hara-kiri is by failing to protect its intellectual property (i.e. knowledge) and/or ignoring the machines of the second industrial revolution. Only those companies which make current machines smarter, or exploit them to their advantage, are set to dominate the coming twenty first century.

The dream of the ultimate intelligent machine which Charniak and McDermott [2] dare mention — ‘...the ultimate goal of AI research (which we are very far from achieving) is to build a person, more humbly an animal’ — is being pursued. Whether it is achieved, unlikely most believe, is immaterial to a large degree. The spin-offs to be gained while striving for it are potentially revolutionary. The field which is attempting to realise the machine intelligence dream is called artificial intelligence (AI).

2. Artificial intelligence — an historical view

Artificial intelligence is still a young discipline, just forty years old. It certainly did not have an easy birth. A pioneer of no lesser status than John Von Neumann (widely regarded as the father of computing) had written, just before his death, a long argument why computers will never exhibit intelligence. Undeterred, a crop of young scientists persisted, and at the now historic Dartmouth College conference of 1956 where several top American researchers came together to discuss the future of machine intelligence, AI was born. Among the researchers were Claude Shannon, Marvin Minsky, John McCarthy (later, the inventor of the LISP programming language), as the main organisers, plus the late Allen Newell and the Nobel laureate, Herbert Simon. The latter four researchers arguably spearheaded AI research for the next two decades until about the mid-1970s.

The general goal of AI is to achieve some degree of ‘intelligence’ in machines. We eschew providing a specific definition of AI in favour of providing (or reminding the reader) of the motivations for doing it, not least because there are so many definitions of it anyway. The reader can derive a definition or definitions from the field’s motivation which include [3]:

- computers are now an accepted part of our lives — researchers want to make them smarter (i.e. extend their current capabilities),
- some AI researchers would like to model ‘intelligence’, mind, cognition, consciousness and emotions — many have come into the field to test the validity of their theories on these by programming computer models using AI techniques,
- AI provides a good intellectual pursuit for researchers,
- of course, money must be a factor — it is no coincidence that the major industrialised nations are investing heavily in AI research, since there are enormous financial rewards and power to be accrued from making machines smarter.

Clearly academics draw their inspiration mainly from the first three while industrialists mainly carry out such work primarily because of the fourth reason.

However, the field has been dogged continually by controversy as to its true achievements. To be fair, AI people have arguably been the inventors of their own misfortunes. For example, during the mid-1960s to the mid-1970s, which has been referred to as the romantic period of AI [4], AI researchers pronounced that computers will soon understand stories and dialogue in natural language. Earlier, Herbert Simon, had predicted that we would have a chess champion computer by the mid-1960s. These goals are still outstanding four decades later. There were other spectacular failures. Consider the translation system, which infamously translated the sentence ‘the spirit is willing but the flesh is weak’ into Russian and then back to English. To the amusement of many AI detractors, it returned the sentence ‘the wine is good, but the meat is rotten’. Therefore, even in the 1970s, the euphoria was short-lived, and, by 1975, it was clear that scaling up toy demonstrators to more substantive real-world domains, provide exponential rather than linear problems, and hence are non-trivial.

You would have thought AI researchers would have learnt from the failures of the 1960s and 70s, but during the 1980s they were culpably at it again — hyping the promise of AI with the help of the media and public. Our point is simple. Such hype generates unrealistic expectations. The exaggerated claims, over-expectations and ‘over-sell’ eventually drown out real (and sometimes excellent) achievements, which never match the hype. The aftermath, in retrospect, was quite predictable — interest in the area waned and, as night follows day, funding cuts followed. In the UK, this was evidenced by Sir James Lighthill’s devastating 1973 report for the then Science and Engineering Council [5], which concluded that AI’s erstwhile promises were dubious. This effectively killed AI research in the UK for a decade.

Furthermore, in such a climate of virtually inelastic claims, it was only normal to expect virulent attacks to come from the critics, who have argued cogently that AI research was an expensive irrelevance. Some critics, notably the Dreyfus brothers [6, 7], and the likes of Searle [8] arguably have, to some degree, flourished in their careers by engaging in wholesale debunking of the claims of AI. Recently, the Oxford scientist, Roger Penrose has joined this list of AI detractors with the publication of his *Shadows of the Mind* [9]. In it, Penrose argues that the brain/mind is a physical reality whose understanding transcends the computational metaphor which is at the heart of AI. He posits that investigations into quantum mechanics, classical and quantum physics, microbiology, etc, may illuminate further our understanding of the workings of mind, brain and consciousness.

In any case, we contend that most AI people do not bother to understand all these philosophical arguments — to them they are sterile. Others have concluded that these critics’ arguments are irrelevancies to the machine

intelligence venture. However, while some of their arguments smack of sophistry, some others are very cogent indeed. These critics, notwithstanding, the dream of machine intelligence lives on.

3. Machine intelligence — some success stories

We split this section into two parts — the first is more general and world-wide in outlook, while the second is more BT-oriented.

3.1 Some general AI success stories

Lest the latter part of the preceding section reads as a *mea culpa* of the failings of AI, we now highlight, only briefly, some of its success stories. AI now encompasses many areas of investigation including knowledge-based systems, natural-language processing, speech recognition and synthesis, automated reasoning (rule-based reasoning, case-based reasoning, model-based reasoning, qualitative and/or spatial reasoning, etc), automatic programming, vision, search and knowledge representation, planning and scheduling, neural networks, robotics, distributed AI, soft computing, etc.

AI is now part of our lives in ways unimaginable just a decade ago. Particularly since about 1990, AI's use is now quite widespread. Cars are assembled in factories by robots which employ AI technology. Indeed, Munakata [10] reports that estimates of the number of industrial robots world-wide is of the order of half a million. Vision applications are used to ensure quality control in several industries; for example, it is used in the area of electronic inspection, particularly of very large scale integration (VLSI) chips and circuit boards. Other industrial uses of vision applications are in stereo for terrain extraction, target detection and tracking, and x-ray inspection of jet engine turbine blades [11].

Many (particularly Japanese) banks now use speech recognition systems to handle several thousand customer calls daily; though these systems have limited vocabularies, they increasingly combine speaker-independent speech-recognition and speech-synthesis technologies. AI technology is being exploited in weather prediction, mining and in predicting the share movements on the stock market. Not so long ago, the chess program, Deep Blue, won a game against the long-reigning world chess champion, Gary Kasparov. Herbert Simon's prediction of having a computer world chess champion may soon become a reality.

Knowledge-based (expert) systems (KBS) are now established in permanent and secure roles in industry as assistants to humans, decision-making components of more complex systems, critics, automatic planners, designers and schedulers [12]. Indeed, some current estimates of the total

dollar value for the expert systems market range from 800 millions (1990) to 1.6 billions (1995). Whatever the real figure is, there is no denying the fact that there has been a real uptake of the technology. One particular KBS, XCON (DEC's configuration expert system), is claimed to perform six times as many functions as the personnel used to perform and saves DEC about 20 million dollars a year (this is a 1984 figure). Indeed, Kurzweil [1] notes that XCON '...reportedly is doing the work of over 300 human experts with substantially higher accuracy', (p 292). More recently, Hayes-Roth and Jacobstein [12] report that XSEL, one of the systems in the XCON suite, reduced a three-hour system configuration/alternative generation task to 15 minutes and also reduced the number of non-manufacturable systems specified from 30% to 1%; these savings are claimed to be worth 70 million dollars annually.

On a domestic front, AI technologies have found their way, and are still finding their way, into domestic appliances like washing machines and microwave ovens. For example, washing machines now possess 'fuzzy logic' chips which can determine what type of fabric is being washed, and can set the correct wash cycle. Some even detect how 'hard' the water is in order to determine how much detergent to use. There are now 'one touch' microwave ovens which work out what to do with most foods placed in them — they are able to work out, for example, whether it is frozen (and thus start by defrosting); they can also work out for how long the defrosting should go on, etc. These fuzzy logic chips make such decisions from information obtained from sensors within these appliances. Other household appliances which contain such technology include refrigerators, vacuum cleaners, rice cookers and air conditioners. Widrow et al [13] report the exploitation of neural networks in domains including pattern classification, prediction and financial analysis, and control and optimisation.

We could go on, but we believe these examples amply demonstrate some of the incontestable contributions AI is having on our lives, despite the fact that true machine intelligence has not yet been achieved.

3.2 Machine intelligence in BT — a brief review and some success stories

BT did not start investigating the area of artificial intelligence in earnest until about 1984 when a few AI-related groups established themselves. The main reasons were twofold — firstly, very little AI work proceeded in the UK at that time, other than in pocket areas such as the universities of Edinburgh, Essex and Sussex, thanks to the Lighthill report mentioned earlier. Secondly, very few industrial research organisations in the UK were investing in AI-related R&D while the majority of the companies were cautiously awaiting the area to gain sufficient

credibility and self-assurance. With some help from the AI and expert systems craze of the mid-1980s and hard work of some internal champions, groups were set up in BT for knowledge-based systems, natural-language processing, speech processing, speech synthesis and recognition, and data mining. Over the years, some of the groups have evolved tremendously, and addressed different areas of AI. For example, the original knowledge-based systems (also known as advanced information processing (AIP)) group, towards the late 1980s and early 1990s, also began investigating domains including machine learning, distributed AI, and planning and scheduling. Currently, this group is called the Intelligent Systems Research (ISR) group, and its researchers are investigating novel technologies including advanced planning and scheduling, software agents, adaptive learning and soft computing.

Why does/should BT strive for machine intelligence? There are several general reasons. We believe AI techniques would contribute towards the following:

- to facilitate BT in providing better-quality, and a wider range of, services as well as providing 'value-added' services to new and existing customers,
- to facilitate BT in managing its vast national and international telecommunications networks and reducing its operational cost,
- to facilitate BT in responding to the needs of its customers and hence not only keeping but attracting new customers,
- to facilitate BT in expanding into the global telecommunications market,
- to provide 'proof of concept' demonstrators that will open exploitation paths within BT — several of which are presented in this issue,
- to seek routes to downstream already-proven ideas to the rest of the BT business,
- to increase BT patents portfolio, its visibility of public domain results and its international R&D reputation,
- to develop core skills in machine intelligence and intelligent software systems.

So far, there have been many success stories within BT; a few recent and nascent ones are highlighted here.

- Intelligent scheduling and resource management — effective resource management has a major impact on BT's telecommunications network performance and profitability, and significantly contributes to the advantages provided to its customers. The ISR group at

BT Laboratories (BTL) has developed intelligent resource management systems capable of providing flexible planning, scheduling and allocation of resources in dynamic environments such as visual broadcast service routing, workforce scheduling, etc. These systems are developed using AI-based technologies such as constraint modelling, satisfaction and optimisation techniques, stochastic search and distributed problem-solving techniques, and constraint logic programming.

- Speech synthesis — BT's speech group has developed the Laureate text-to-speech system [14], which enables customers to access easily text-based information, anywhere, anytime, using an ordinary fixed or mobile telephone. Its performance betters that of many commercial speech-synthesis systems; at normal listening levels, it betters any other system against which it has been tested. It has been easily integrated into a number of different systems which include telephone access to e-mail, road traffic congestion, weather forecast, WWW pages and medical information lines. It can be found on BT's external Web server¹.
- Text summarisation — we are living in the era of information overload, and it will only get worse. The language group at BTL has developed Netsumm² — a text summariser that accepts plain-text documents, and automatically picks out the 'most important' parts of the text, e.g. by highlighting them. Alternatively, it can extract the sentences from the text to produce an abridgement of the article. The results of this prototype, which is still being evaluated, are so far very encouraging.
- Movie recommendation — the ISR group has developed MORSE, an agent which provides personalised movie recommendation. It works by first asking the user to rate as many films he/she has seen as possible. Using this data, MORSE recommends films which others with 'similar tastes' have rated highly. MORSE's performance so far equals and even betters other similar systems (see Fisk [15], and BT's WWW server³).

It is perhaps useful to note that special issues of the BT Technology Journal on some of the more established AI sub-areas mentioned earlier, relative to soft computing and software agents, already exist. Some recent issues with AI-related themes or which include a significant number of AI-oriented papers include the following:

¹ <http://www.labs.bt.com/innovate/speech/laureate/index.htm>

² <http://www.labs.bt.com/innovate/informat/netsumm/index.htm>

³ <http://www.labs.bt.com/innovate/multimed/morse/morse.htm>

- Network Management, BT Technology Journal, Volume 9, Number 3, July 1991,
- Artificial Neural Networks, BT Technology Journal, Volume 10, Number 3, July 1992,
- Modelling Change in Telecommunications, BT Technology Journal, Volume 12, Number 2, April 1994,
- Computer-Aided Decision Support, BT Technology Journal, Volume 12, Number 4, October 1994,
- Advanced Information Processing Techniques for Resource Scheduling and Planning, BT Technology Journal, Volume 13, Number 1, January 1995,
- Speech Technology for Telecommunications, BT Technology Journal, Volume 14, Number 1, January 1996.

4. Conclusions

This paper has provided a brief, historical introduction, for the uninitiated, to machine intelligence and AI. It is vital that such domains are investigated because the immense complexity of modern communications systems is forcing BT to seek novel and powerful solutions to meet the challenges of tomorrow's systems. Furthermore, as the convergence of telecommunications and computing gathers pace the difficulties facing global operators such as BT will grow. And this growth in complexity is not set to abate in the next few years.

Artificial intelligence, still a relatively young scientific and engineering discipline, has been (and is still being) successfully applied by BT researchers and systems developers in order to provide solutions to some of the problems envisaged.

Much of the AI research and intelligent systems engineering required to deal with the grand challenges ahead is still outstanding.

References

- 1 Kurzweil R: 'The Age of Intelligent Machines', Cambridge, Massachusetts, MIT Press (1992).
- 2 Charniak E and McDermott D: 'Introduction to Artificial Intelligence', Cambridge, Massachusetts Addison-Wesley (1982).
- 3 Nwana H S: 'Introducing Artificial Intelligence — a gentler and less formal approach', Unpublished manuscript, BT Laboratories (1996).
- 4 Jackson P: 'Introduction to Expert Systems', London, Addison-Wesley (1990).
- 5 Lighthill J: 'UK Science and Engineering Report', (1973).
- 6 Dreyfus H L: 'What computers can't do', New York, Harper & Row (1972).
- 7 Dreyfus H L: 'What computers still can't do: a critic of artificial reason', New York, Harper & Row (1992).
- 8 Searle J: 'Is the brain's mind a computer program? no: a program merely manipulates symbols whereas a brain attaches meaning to them', Scientific American, No 1, pp 20-25 (1990).
- 9 Penrose R: 'Shadows of the Mind: A search for the missing science of consciousness', Oxford University Press (1994).
- 10 Munakata T: 'Commercial and Industrial AI', Communications of the ACM, 37, No 3, pp 23-25 (1994).
- 11 Grimson W E L and Mundy J L: 'Computer Vision Applications', Communications of the ACM, 37, No 3, pp 45-51 (1994).
- 12 Hayes-Roth F and Jacobstein J: 'The State of Knowledge-Based Systems', Communications of the ACM, 37, No 3, pp 27-39 (1994).
- 13 Widrow B, Rumelhart D E and Lehr M A 'Neural Networks: applications in Industry, Business and Science', Communications of the ACM, 37, No 3, pp 93-105 (1994).
- 14 Page J H and Breen A P: 'The Laureate text-to-speech system — architecture and applications', BT Technology J, 14, No 1, pp 57—67 (January 1996).
- 15 Fisk D: 'An application of social filtering to movie recommendation', BT Technol J, 14, No 4, pp 124—132 (October 1996).

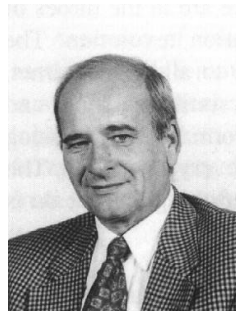


Hyacinth S Nwana holds a BSc in Computer Science and Electronic Engineering (1985), an MSc in Computer Science (1986) and a PhD in the area of Artificial Intelligence (1989). Between 1989 and 1995, he worked in various roles (lecturer, researcher, contract and visiting scientist) for the Universities of Liverpool, Keele, Calgary (sponsored by the Royal Society), Shell Research Labs and BT Laboratories.

He joined BT in October 1995 and leads part of BT's software agents research programme, as well as a team researching into the engineering of agent-based systems, and is contactable on e-mail at: hyacinth@info.bt.co.uk. He has a broad range of AI-related interests and has written on software agents, AI and education, distributed AI, machine learning, neural networks, knowledge acquisition, computer science education and cognitive science. In 1991, he principally won the Digital Equipment Corporation (DEC) European Artificial Intelligence prize. He is currently pursuing a part-time MEd degree at Cambridge. He is a member of the British Computer Society and a Chartered Engineer.



Nader Azarmi is the technical group leader of the Intelligent Systems Research group at BT Laboratories. He joined BT in December 1989, working on the application of AI techniques to telecommunications network automatic fault diagnosis. He went on to set up an extensive research and development programme in the application of AI planning and scheduling technology to BT's resource management problems. He is currently managing an international AI and Machine Intelligence research programme. He received a BSc Hons in Computer Science and Mathematics from the University of Manchester, an MSc in Artificial Intelligence and a MPhil in Computing Science from Essex university. He is currently completing a PhD in Computer Science at Imperial College, London University. Prior to joining BT he held a senior lectureship position in the Computer Science and Electronics School of Kingston University.



Robin Smith has devoted his career to the engineering aspects of telecommunications systems. During the past 12 years he was head of a research unit at BT Laboratories which investigated the application of AI techniques to many areas of system management and control.

Currently he manages his own company (TelicTech Ltd), and is a visiting Professor at Imperial College and a member of the EPSRC Computing College. He is a Chartered Engineer and Fellow of the IEE.